Dilip Bhakadiwal

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TECHNICAL SKILLS

Agentic AI & Orchestration: LangGraph, LangChain, Model Context Protocol (MCP), ReAct Agent Workflows
Backend & Cloud Pipelines: FastAPI, AWS (App Runner, S3, CloudFront), Docker, Render, GitHub Actions (CI/CD)
Data Processing & Retrieval: Pinecone Serverless, PostgreSQL (Aiven), FastEmbed, GraphRAG, Neo4j, Hybrid Retrieval
LLM Integration & Evals: OpenRouter APIs, Pytest, LLM Evals, Prompting, Langsmith, Llamaindex
Programming & Tools: Python, Pydantic, Git, Pandas, NumPy, Scikit-learn

EDUCATION

Defence Institute of Advanced Technology (DIAT) — 2024 – 2026
Master of Technology in Artificial Intelligence – CGPA: 7.33 — Pune, Maharashtra

MBM University — 2019 – 2023
Bachelor of Engineering in Electronics and Computer Engineering — Jodhpur, Rajasthan

KEY AI ENGINEERING PROJECTS

MarketPulse AI: Agentic Financial Terminal | *LangGraph, FastAPI, PostgreSQL, SSE, Render* — 2026

- Developed a production-grade conversational financial terminal using LangGraph and FastAPI, integrating a self-healing 7-stage web scraper pipeline with automatic symbol alias resolution.
- Implemented real-time Server-Sent Events (SSE) token-streaming and a dynamic UI explorer, backed by an Aiven Managed PostgreSQL database handling advanced algorithmic SQL synthesis (Z-score anomaly detection).
- Engineered AST-level SQL security guardrails to block destructive queries and deployed via automated CI/CD on Render with a resilient 3-path cloud price fallback.

Enterprise RAG Pipeline (Redwood Inference) | *LangGraph, Pinecone, Groq, FastEmbed* — 2026

- Architected a highly cost-optimized, automated RAG pipeline over the EnterpriseRAG-Bench dataset using LangGraph and FastAPI.
- Integrated FastEmbed and a Pinecone Serverless vector database for retrieval, utilizing the Groq API (Llama 3.3 70B) for synthesis alongside intelligent source deduplication.
- Containerized the application via Docker, optimizing image builds with pre-cached model weights to eliminate cold-start latency, and established a GitHub Actions CI/CD pipeline gated by sequential Ragas evaluations.

RESEARCH EXPERIENCE

Edge AI Object Detection System on FPGA and Jetson Platforms — Sept. 2025 – Jan. 2026
Research Project, DIAT — Pune, Maharashtra

- Designed a custom lightweight YOLOv8n architecture optimized for embedded edge deployment and real-time object detection.
- Performed model quantization from FP32 to INT8, reducing model size significantly and improving inference efficiency on edge hardware.
- Deployed the quantized model on a Xilinx FPGA accelerator, achieving approximately 13 FPS real-time inference.
- Evaluated the FP32 model on an NVIDIA Jetson Orin (2048 CUDA cores), achieving 45 FPS real-time object detection.
- Integrated a lightweight LLaMA 1B model to generate contextual natural-language descriptions of detected objects in real time.

Focal-CBAM Fish-YOLO | *Funded by MoES, Govt. of India* — Aug. 2025 – Oct. 2025
Research Project, DIAT — Pune, Maharashtra

- Developed an attention-enhanced YOLOv8 detection architecture using a novel Focal-CBAM module to improve feature attention in underwater environments.
- Conducted experiments on the RUOD underwater object detection dataset, improving detection performance under low-visibility and noisy conditions.
- Optimized multi-scale detection heads and feature extraction pipelines; outperformed baseline YOLO architectures on underwater localization benchmarks.
- Manuscript accepted at the ICASA 2025 Conference (icasa-conf.co.uk); publication pending.

PUBLICATIONS

Deep Underwater Fish Detection via Focal Modulated Channel Attention in YOLO — Accepted

- *First Author* | **ICASA Conference 2025** | Funded by **MoES, Govt. of India**

ANIMA: YOLOv8-Based Framework for Object Detection & Compression in Satellite Imagery — Published

- *Second Author* | **IEEE Xplore**, IEEE Pune Section | doi:10.1109/...11379260